GRADE 12 CHEMISTRY

UNIT 5 — ENVIRONMENTAL CHEMISTRY

Environmental components and cycles, pollution, climate change and green chemistry

Exam-focused revision booklet

Prepared from the uploaded Ethiopian Grade 12 Chemistry Student Textbook (Ministry of Education, 2023). The explanations below are newly written study notes, not copied textbook pages. Diagrams are original schematic summaries.

How to use this booklet

Read the core idea, memorize the key relationship or equation, redraw the visual summary from memory, then answer both alternative questions without looking. For calculations, show units and working. For process questions, give the stages in the correct order.

CORE CONCEPTS	KEY EQUATIONS / PROCESSES	EXAM PRACTICE

Coverage: 19 focused topic sheets plus this guide page = 20 pages minimum. The two questions on each topic are alternatives for self-testing.

5.01 Environmental chemistry

Core concept

Environmental chemistry studies the production, transport, reactions, effects and fate of chemical species in air, water, soil and living systems, including effects of human activities. It is multidisciplinary because it connects chemistry with biology, geology, physics and environmental science. Follow a pollutant from its source through transport, transformation and final fate.

Remember

Source → transport → reaction → effect → fate

Environmental chemistry	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define environmental chemistry in your own words.

B. Why is environmental chemistry considered multidisciplinary?

5.01 • Exam diagram & MCQ practice

Key exam idea: Environmental chemistry studies chemical substances, reactions, movement and effects in air, water, soil and living systems.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Environmental chemistry is multidisciplinary because it:

A. connects chemistry with biology, geology and other sciences
B. studies only laboratory glassware
C. excludes human activities
D. focuses only on metals

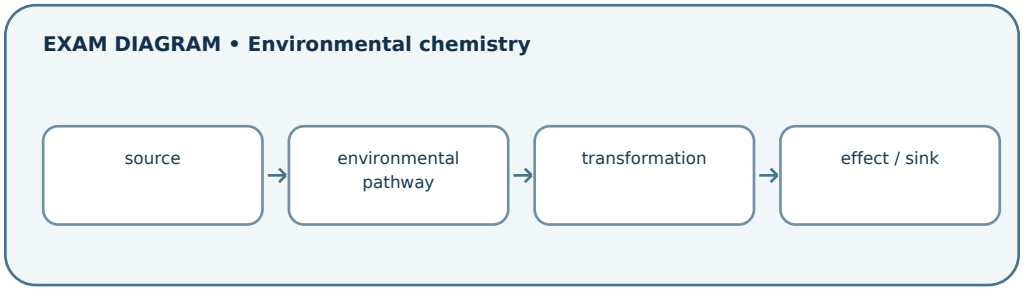
Answer: A. Environmental problems involve chemical processes interacting with organisms, Earth systems and human activity.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.02 Components of the environment

Core concept

The environment includes interacting atmosphere, hydrosphere, lithosphere and biosphere. Chemical species move between these components through natural cycles and human activities. An environmental question may ask for examples, interactions or chemical processes in each sphere. Remember that a change in one component can affect the others.

Remember

Atmosphere • hydrosphere • lithosphere • biosphere

Components of the environment	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name the major components of the environment.

B. Give one example of how a chemical substance can move between two environmental components.

5.02 • Exam diagram & MCQ practice

Key exam idea: The four interacting environmental spheres are atmosphere, hydrosphere, lithosphere and biosphere.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which sphere includes Earth's water?

A. hydrosphere
B. atmosphere
C. lithosphere
D. biosphere

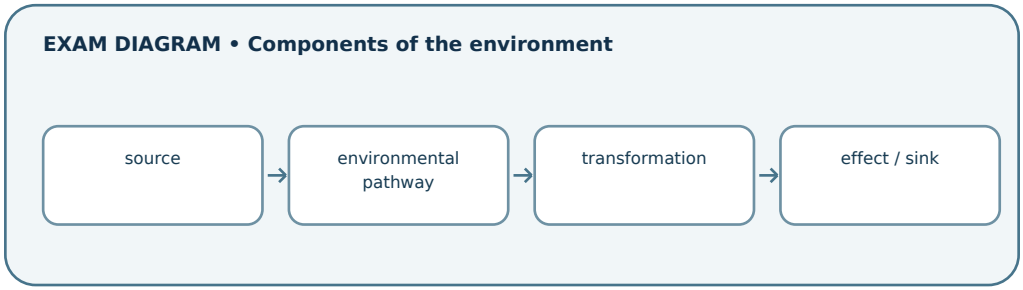
Answer: A. The hydrosphere includes oceans, rivers, lakes, groundwater and other water stores.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.03 Pollutants, contaminants and sinks

Core concept

A pollutant is a substance or agent present at a level that causes harmful effects. A contaminant is an unwanted substance; whether it causes harm depends on its concentration and context. A sink is a reservoir or process that removes a substance from a particular environmental compartment. The textbook also introduces BOD and threshold limit value.

Remember

Pollutant source → environmental compartment → sink

Pollutants, contaminants and sin	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Distinguish pollutant from contaminant.

B. What is an environmental sink? Give one example.

5.03 • Exam diagram & MCQ practice

Key exam idea: A pollutant causes harm at its level; an environmental sink stores or removes a substance from active circulation.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is an example of an environmental sink for carbon dioxide?

A. ocean water
B. a car exhaust pipe
C. a factory chimney
D. a fuel tank before combustion

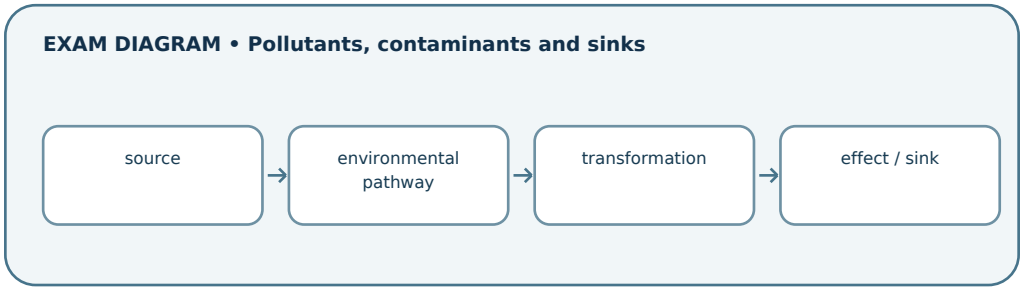
Answer: A. The ocean absorbs and stores a portion of atmospheric CO_2, acting as a carbon sink.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.04 Biological oxygen demand (BOD)

Core concept

Biological oxygen demand is related to the oxygen consumed by microorganisms as they decompose biodegradable organic matter in water. High BOD indicates a large oxygen demand and can reduce dissolved oxygen available to aquatic organisms. Use BOD as an indicator when discussing organic water pollution and wastewater quality.

Remember

High biodegradable load → high BOD → lower dissolved oxygen

Biological oxygen demand (BOD)	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. What does BOD indicate in a water sample?

B. Explain how high BOD can affect aquatic life.

5.04 • Exam diagram & MCQ practice

Key exam idea: BOD measures the oxygen microorganisms consume while decomposing biodegradable organic matter in water.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A high BOD value usually indicates:

A. high biodegradable organic pollution
B. very pure water with no organic matter
C. zero microbial activity
D. high dissolved oxygen at all times

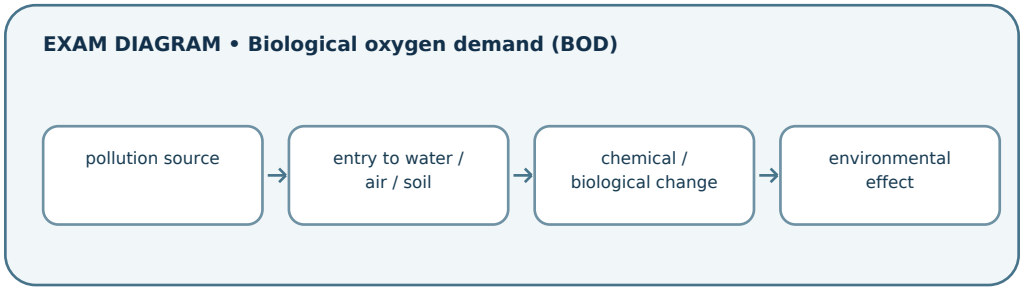
Answer: A. More biodegradable waste supports greater microbial decomposition and oxygen consumption, giving a higher BOD.

Question 2. In the diagram, which labelled step comes immediately after “pollution source”?

A. entry to water / air / soil
B. chemical / biological change
C. environmental effect
D. pollution source

Answer: A. The schematic is arranged in sequence: pollution source → entry to water / air / soil → chemical / biological change → environmental effect.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.05 Threshold limit value

Core concept

A threshold limit value is a reference limit for exposure to a substance, used in assessing whether exposure is within an acceptable occupational or environmental guideline. The textbook introduces this term among environmental chemistry concepts. When answering, state that it is a limit/reference value and relate it to exposure and risk rather than treating it as a chemical concentration formula.

Remember

TLV = exposure reference limit

Threshold limit value	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define threshold limit value (TLV).

B. Why are exposure limits useful in environmental and workplace protection?

5.05 • Exam diagram & MCQ practice

Key exam idea: A threshold limit value is an occupational exposure guideline for the concentration of a substance over a specified period.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The main purpose of a threshold limit value is to:

A. guide control of exposure to hazardous substances
B. measure the pH of rivers
C. determine polymer molar mass
D. calculate reaction enthalpy

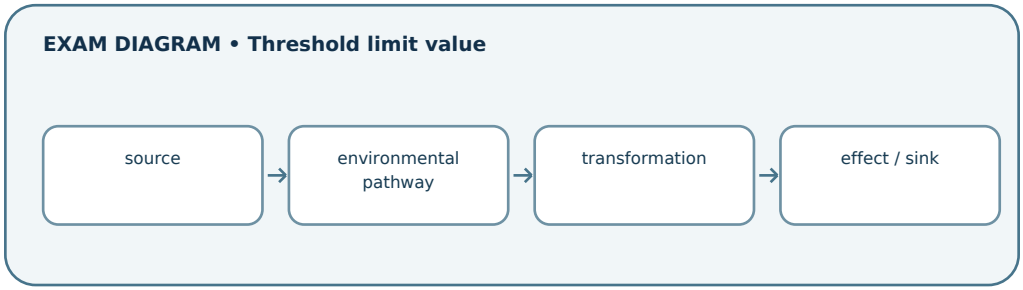
Answer: A. Exposure limits help assess and reduce the risk of harmful workplace exposure.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.06 Hydrological cycle

Core concept

The hydrological cycle describes continuous movement of water through evaporation, transpiration, condensation, precipitation, infiltration, runoff and collection. Solar energy drives evaporation, and gravity returns water to the surface. Water also transports dissolved substances, so the cycle influences the movement and distribution of chemicals.

Remember

Evaporation → condensation → precipitation → runoff/infiltration

Hydrological cycle	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Describe the main stages of the hydrological cycle.

B. How can the water cycle transport pollutants?

5.06 • Exam diagram & MCQ practice

Key exam idea: Water moves through evaporation, transpiration, condensation, precipitation, infiltration and runoff.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which process changes liquid water into water vapour?

A. evaporation
B. condensation
C. precipitation
D. infiltration

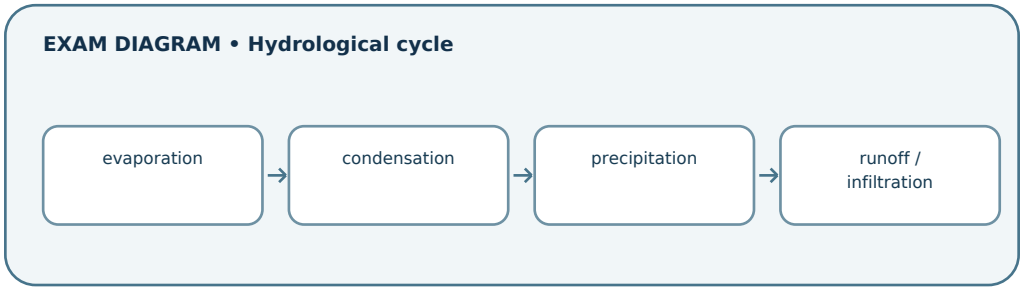
Answer: A. Evaporation transfers water from liquid surfaces into the atmosphere as vapour.

Question 2. In the diagram, which labelled step comes immediately after “evaporation”?

A. condensation
B. precipitation
C. runoff / infiltration
D. evaporation

Answer: A. The schematic is arranged in sequence: evaporation → condensation → precipitation → runoff / infiltration.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.07 Carbon cycle

Core concept

Carbon moves among the atmosphere, organisms, water and geological materials. Photosynthesis removes carbon dioxide from the atmosphere and incorporates carbon into organic matter; respiration, decomposition and combustion return carbon dioxide. Human activities such as fossil-fuel combustion alter the balance of carbon flows.

Remember

CO■ → photosynthesis → biomass → respiration/decomposition/combustion

Carbon cycle	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain the role of photosynthesis in the carbon cycle.

B. Name two processes that return CO■ to the atmosphere.

5.07 • Exam diagram & MCQ practice

Key exam idea: Photosynthesis removes CO_2 from air or water; respiration, decomposition and combustion return carbon to the environment.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which process removes atmospheric CO_2?

A. photosynthesis
B. combustion
C. respiration
D. decomposition

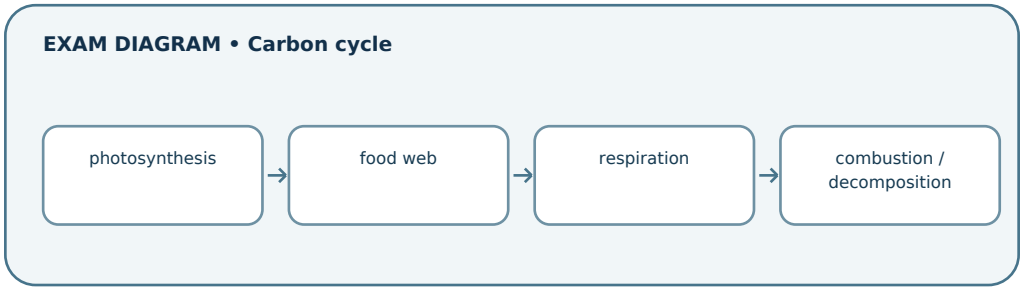
Answer: A. Plants and other photosynthetic organisms take in CO_2 to build organic molecules.

Question 2. In the diagram, which labelled step comes immediately after “photosynthesis”?

A. food web
B. respiration
C. combustion / decomposition
D. photosynthesis

Answer: A. The schematic is arranged in sequence: photosynthesis → food web → respiration → combustion / decomposition.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.08 Nitrogen cycle

Core concept

The nitrogen cycle moves nitrogen among the atmosphere, soil and organisms. Important processes include nitrogen fixation, nitrification, assimilation, ammonification and denitrification. Different microorganisms participate in transformations between nitrogen-containing compounds. Learn the order, the chemical form involved and the role of organisms in each stage.

Remember

N■ → fixed nitrogen → plant uptake → decomposition → N■

Nitrogen cycle	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name the main processes in the nitrogen cycle.

B. What is the role of denitrifying bacteria?

5.08 • Exam diagram & MCQ practice

Key exam idea: The nitrogen cycle includes fixation, nitrification, assimilation, ammonification and denitrification.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which process converts nitrate into nitrogen gas and returns it to the atmosphere?

A. denitrification
B. nitrification
C. assimilation
D. photosynthesis

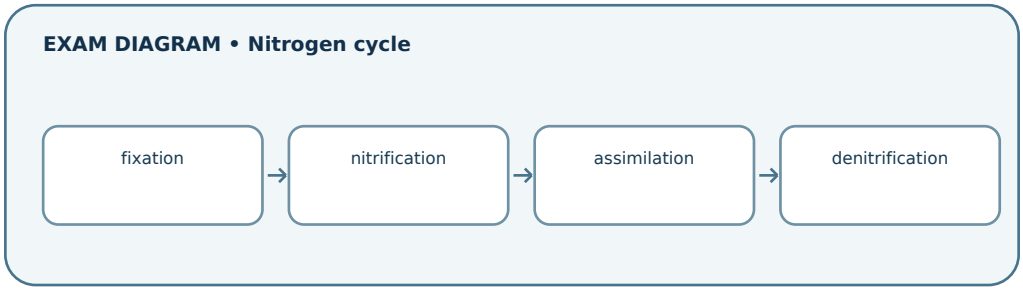
Answer: A. Denitrifying bacteria reduce nitrate to gaseous nitrogen compounds, often N_2.

Question 2. In the diagram, which labelled step comes immediately after “fixation”?

A. nitrification
B. assimilation
C. denitrification
D. fixation

Answer: A. The schematic is arranged in sequence: fixation → nitrification → assimilation → denitrification.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.09 Oxygen cycle

Core concept

Oxygen circulates through the atmosphere, hydrosphere and biosphere. Photosynthesis releases oxygen, while respiration, decomposition and combustion consume it. Oxygen also participates in oxidation reactions in air, water and soil. Relate oxygen-cycle processes to the carbon cycle because photosynthesis and respiration connect both.

Remember

Photosynthesis releases O■; respiration and combustion use O■

Oxygen cycle	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name two processes that release oxygen and two that consume it.

B. Explain one connection between the oxygen and carbon cycles.

5.09 • Exam diagram & MCQ practice

Key exam idea: Photosynthesis releases oxygen; respiration, combustion and oxidation consume oxygen.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which process releases molecular oxygen into the atmosphere?

A. photosynthesis
B. aerobic respiration
C. combustion
D. rusting

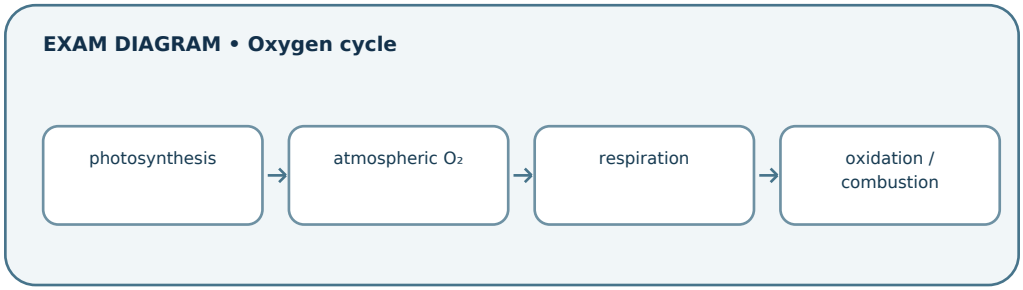
Answer: A. Photosynthesis splits water and releases O_2 as a product.

Question 2. In the diagram, which labelled step comes immediately after “photosynthesis”?

A. atmospheric O_2
B. respiration
C. oxidation / combustion
D. photosynthesis

Answer: A. The schematic is arranged in sequence: photosynthesis → atmospheric O_2 → respiration → oxidation / combustion.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.10 Sulfur cycle

Core concept

Sulfur moves through rocks, soil, water, organisms and the atmosphere. Natural and human processes can release sulfur compounds; combustion of sulfur-containing fuels can produce sulfur dioxide. Atmospheric transformations and deposition return sulfur to land and water. Sulfur dioxide is important in discussions of air pollution and acid deposition.

Remember

Sulfur-containing fuel + O■ → SO■

Sulfur cycle	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. How can combustion contribute to the sulfur cycle?

B. Explain why sulfur dioxide is an environmental concern.

5.10 • Exam diagram & MCQ practice

Key exam idea: Sulfur moves through rocks, soil, water, organisms and air; combustion of sulfur-containing fuels can release SO_2.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Burning sulfur-containing coal can release:

A. sulfur dioxide
B. pure nitrogen
C. helium
D. ozone only

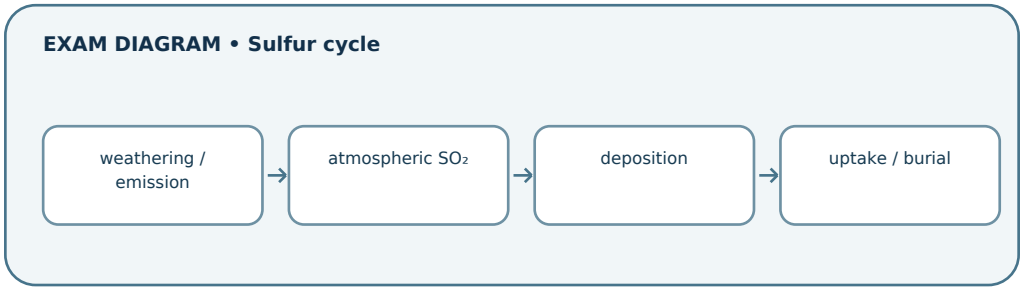
Answer: A. Sulfur in the fuel is oxidized during combustion, producing sulfur dioxide.

Question 2. In the diagram, which labelled step comes immediately after “weathering / emission”?

A. atmospheric SO_2
B. deposition
C. uptake / burial
D. weathering / emission

Answer: A. The schematic is arranged in sequence: weathering / emission → atmospheric SO_2 → deposition → uptake / burial.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.11 Phosphorus cycle

Core concept

Phosphorus moves mainly through rocks, soil, water and living organisms. Weathering releases phosphate, plants absorb it, animals obtain it through food, and decomposition returns it to the environment. Unlike carbon and nitrogen cycles, the textbook emphasizes that the phosphorus cycle has no major gaseous atmospheric phase.

Remember

Rock weathering → phosphate → plants → animals → decomposition

Phosphorus cycle	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Outline the main stages of the phosphorus cycle.

B. Why is the phosphorus cycle described as lacking a major gaseous phase?

5.11 • Exam diagram & MCQ practice

Key exam idea: Phosphorus cycles mainly through rocks, soil, water and organisms and has no major atmospheric gaseous phase.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why is the phosphorus cycle described as mainly sedimentary?

A. it has no major gaseous atmospheric phase
B. phosphorus exists only as a gas
C. it occurs only in the ocean
D. it is driven only by sunlight

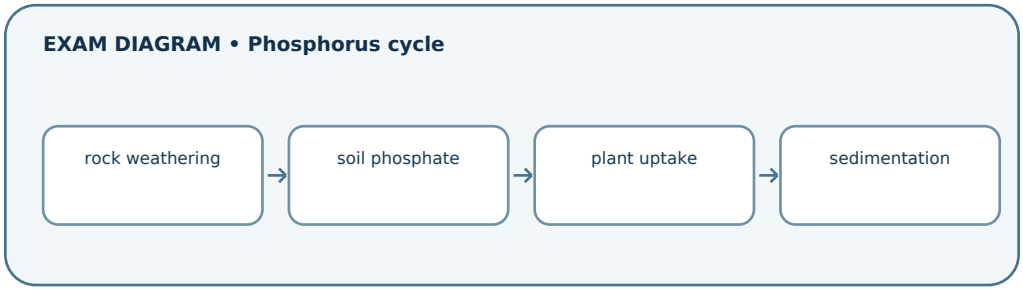
Answer: A. Phosphorus is mainly transferred through phosphate in rocks, soil, water and organisms rather than a large gas reservoir.

Question 2. In the diagram, which labelled step comes immediately after “rock weathering”?

A. soil phosphate
B. plant uptake
C. sedimentation
D. rock weathering

Answer: A. The schematic is arranged in sequence: rock weathering → soil phosphate → plant uptake → sedimentation.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.12 Environmental pollution

Core concept

Pollution occurs when substances or energy introduced into the environment cause harmful changes. The textbook organizes pollution into air, water and land pollution. For each type, identify sources, pollutants, effects and prevention or control. Distinguish the pollutant itself from the environmental impact it produces.

Remember

Source → pollutant → exposure → environmental effect

Environmental pollution	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define environmental pollution and classify its main forms.

B. Give one source and one effect for each major type of pollution.

5.12 • Exam diagram & MCQ practice

Key exam idea: Pollution occurs when substances or energy cause harmful environmental change; major forms include air, water and land pollution.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is an example of water pollution?

A. untreated sewage entering a river
B. tree planting on degraded land
C. rainwater infiltrating soil
D. photosynthesis in a forest

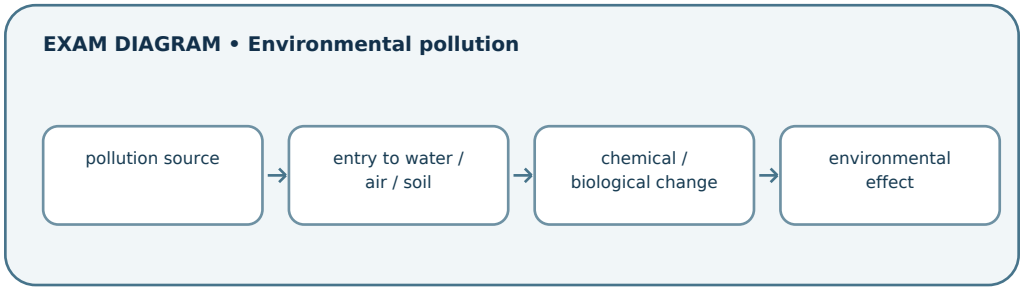
Answer: A. Untreated sewage introduces organic matter and pathogens into water, degrading its quality.

Question 2. In the diagram, which labelled step comes immediately after “pollution source”?

A. entry to water / air / soil
B. chemical / biological change
C. environmental effect
D. pollution source

Answer: A. The schematic is arranged in sequence: pollution source → entry to water / air / soil → chemical / biological change → environmental effect.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.13 Air pollution

Core concept

Air pollution involves harmful substances in the atmosphere. Sources include combustion and industrial activities. Pollutants discussed in the unit include gases and particulates; sulfur dioxide and carbon-containing gases are important examples. Effects may include respiratory harm, damage to vegetation and environmental changes. Prevention includes cleaner fuels, emission control and process improvements.

Remember

Combustion emissions → atmospheric pollutants

Air pollution	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name common sources of air pollution and give examples of pollutants.

B. Explain how sulfur dioxide can affect the environment.

5.13 • Exam diagram & MCQ practice

Key exam idea: Air pollutants include particulate matter, SO_2, NO_x, CO and ground-level ozone from combustion and industrial sources.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Incomplete combustion of carbon-containing fuel can produce:

A. carbon monoxide
B. sodium chloride
C. calcium carbonate
D. pure oxygen

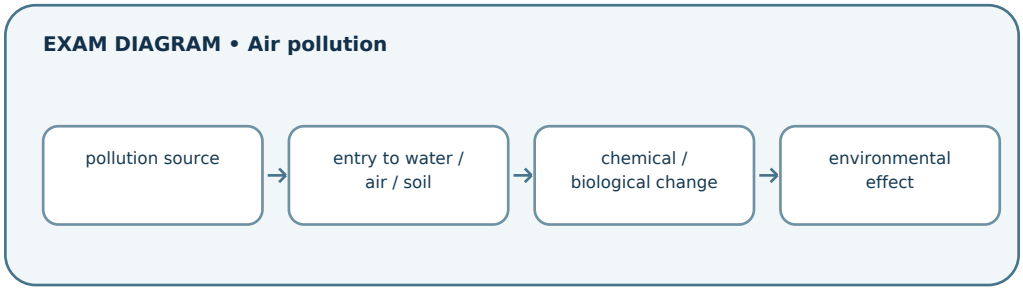
Answer: A. Insufficient oxygen can cause carbon to form CO rather than being fully oxidized to CO_2.

Question 2. In the diagram, which labelled step comes immediately after “pollution source”?

A. entry to water / air / soil
B. chemical / biological change
C. environmental effect
D. pollution source

Answer: A. The schematic is arranged in sequence: pollution source → entry to water / air / soil → chemical / biological change → environmental effect.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.14 Water pollution

Core concept

Water pollution results when unwanted substances enter surface water or groundwater. Sources include domestic wastewater, industrial discharge and agricultural runoff. Organic waste can raise BOD and lower dissolved oxygen; other contaminants may be toxic or cause nutrient enrichment. Prevention involves wastewater treatment, proper waste management and reducing pollutant release at source.

Remember

Organic waste → microbial decomposition → increased BOD

Water pollution	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain the relationship between organic pollution and BOD.

B. List three sources of water pollution and one control measure.

5.14 • Exam diagram & MCQ practice

Key exam idea: Organic waste can raise BOD; nutrients, pathogens, toxic chemicals and sediments are other important water pollutants.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why can untreated organic sewage reduce dissolved oxygen in water?

A. microorganisms consume oxygen during decomposition
B. sewage converts oxygen into nitrogen instantly
C. it stops all microbial activity
D. it increases photosynthesis in every condition

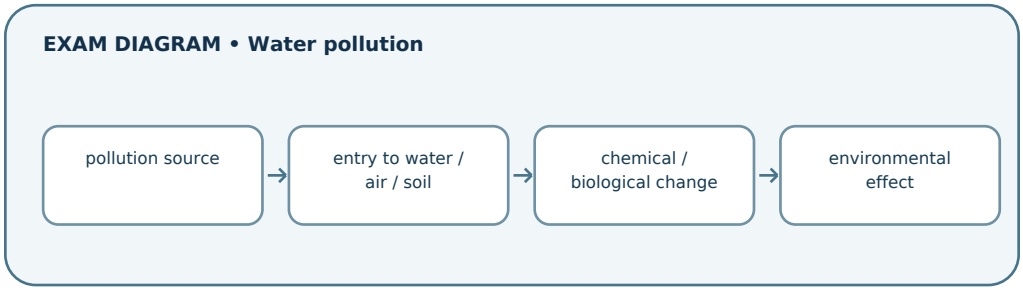
Answer: A. Decomposer microorganisms use dissolved oxygen as they break down organic matter.

Question 2. In the diagram, which labelled step comes immediately after “pollution source”?

A. entry to water / air / soil
B. chemical / biological change
C. environmental effect
D. pollution source

Answer: A. The schematic is arranged in sequence: pollution source → entry to water / air / soil → chemical / biological change → environmental effect.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.15 Land pollution

Core concept

Land pollution occurs when solid waste, chemicals or other harmful materials contaminate soil and land. Sources include improper disposal of household, industrial and agricultural waste. Effects may include reduced soil quality, contamination of groundwater and harm to organisms. Prevention includes waste reduction, reuse, recycling and safe disposal.

Remember

Waste generation → soil contamination → ecological effects

Land pollution	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Identify major causes of land pollution.

B. Explain how improper disposal of chemicals can affect soil and groundwater.

5.15 • Exam diagram & MCQ practice

Key exam idea: Land pollution can result from unmanaged solid waste, hazardous chemicals, mining residues and excessive agrochemical use.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which practice best helps prevent land pollution?

A. safe waste segregation and disposal
B. dumping batteries in open fields
C. releasing chemicals onto soil
D. burning mixed waste in streets

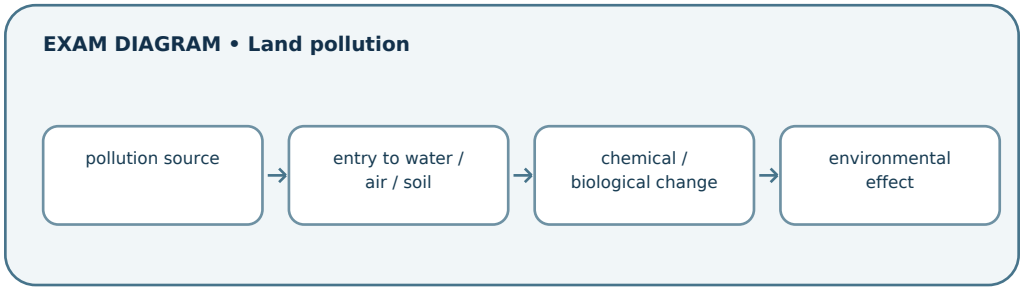
Answer: A. Segregation and proper disposal reduce the release of hazardous materials into soil and groundwater.

Question 2. In the diagram, which labelled step comes immediately after “pollution source”?

A. entry to water / air / soil
B. chemical / biological change
C. environmental effect
D. pollution source

Answer: A. The schematic is arranged in sequence: pollution source → entry to water / air / soil → chemical / biological change → environmental effect.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.16 Global warming and climate change

Core concept

Global warming refers to the increase in Earth's average temperature, while climate change includes broader long-term changes in climate patterns. The textbook links these changes to greenhouse gases and human activities. Distinguish the terms, identify causes and describe effects without treating every weather event as proof of climate change.

Remember

Greenhouse-gas increase → enhanced heat trapping

Global warming and climate chang	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Differentiate global warming from climate change.

B. Give two possible effects of climate change discussed in environmental science.

5.16 • Exam diagram & MCQ practice

Key exam idea: Global warming is the rise in average global temperature; climate change includes broader long-term shifts in climate patterns.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which statement is correct?

A. global warming is one aspect of climate change
B. climate change means only daily weather
C. global warming is unrelated to greenhouse gases
D. the terms refer only to volcanic eruptions

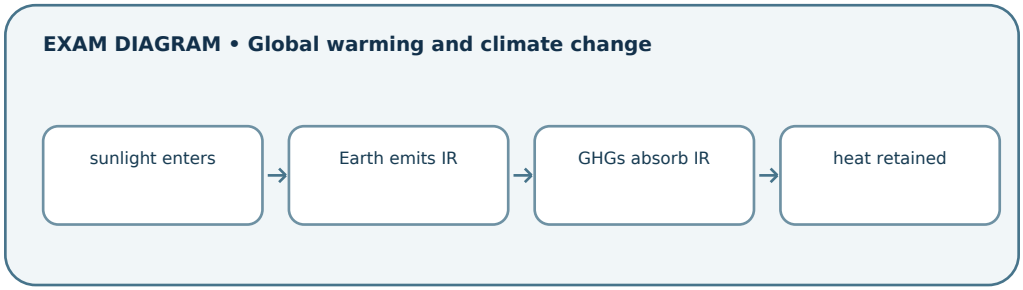
Answer: A. Global warming describes temperature rise, while climate change includes wider changes such as rainfall and extremes.

Question 2. In the diagram, which labelled step comes immediately after “sunlight enters”?

A. Earth emits IR
B. GHGs absorb IR
C. heat retained
D. sunlight enters

Answer: A. The schematic is arranged in sequence: sunlight enters → Earth emits IR → GHGs absorb IR → heat retained.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.17 Greenhouse gases and their chemistry

Core concept

Greenhouse gases absorb and re-emit infrared radiation, contributing to heat retention in the atmosphere. The textbook discusses the chemistry of greenhouse gases and their effects on climate. Carbon dioxide is linked to combustion and carbon-cycle changes; other gases can also contribute. Explain the mechanism clearly: increased greenhouse-gas concentration strengthens the greenhouse effect.

Remember

Greenhouse gases absorb and re-emit infrared radiation

Greenhouse gases and their chemi	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain how greenhouse gases contribute to warming.

B. Name sources of carbon dioxide and describe how human activity can increase its atmospheric level.

5.17 • Exam diagram & MCQ practice

Key exam idea: Greenhouse gases absorb and re-emit infrared radiation, reducing the rate at which Earth loses heat to space.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which gas is a major human-influenced greenhouse gas?

A. carbon dioxide
B. argon
C. neon
D. helium

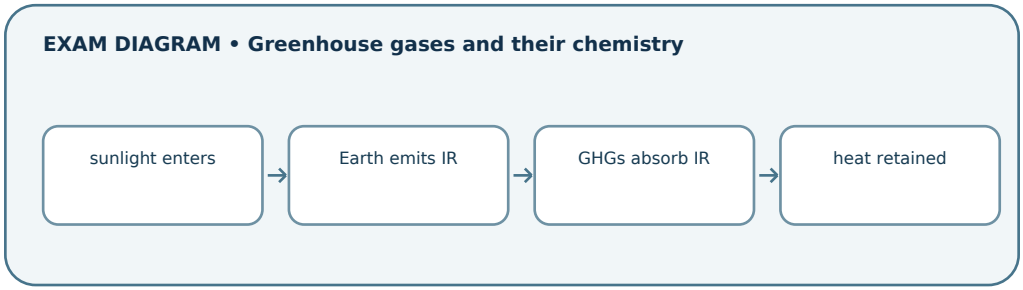
Answer: A. CO_2 absorbs infrared radiation and its atmospheric concentration increases through fossil-fuel use and deforestation.

Question 2. In the diagram, which labelled step comes immediately after “sunlight enters”?

A. Earth emits IR
B. GHGs absorb IR
C. heat retained
D. sunlight enters

Answer: A. The schematic is arranged in sequence: sunlight enters → Earth emits IR → GHGs absorb IR → heat retained.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.18 Green chemistry and its purpose

Core concept

Green chemistry aims to design chemical products and processes that reduce or eliminate hazardous substances and waste. It considers the whole life cycle, from raw materials and synthesis to product use and disposal. It is different from simply cleaning pollution after it has been created: the goal is prevention by design.

Remember

Prevent pollution at source rather than clean it up later

Green chemistry and its purpose	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define green chemistry and explain its purpose.

B. How does green chemistry differ from end-of-pipe pollution treatment?

5.18 • Exam diagram & MCQ practice

Key exam idea: Green chemistry designs products and processes to reduce or eliminate hazardous substances and waste at the source.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Green chemistry primarily aims to:

A. prevent pollution through safer chemical design
B. treat all waste only after it is produced
C. increase hazardous solvent use
D. maximize waste generation

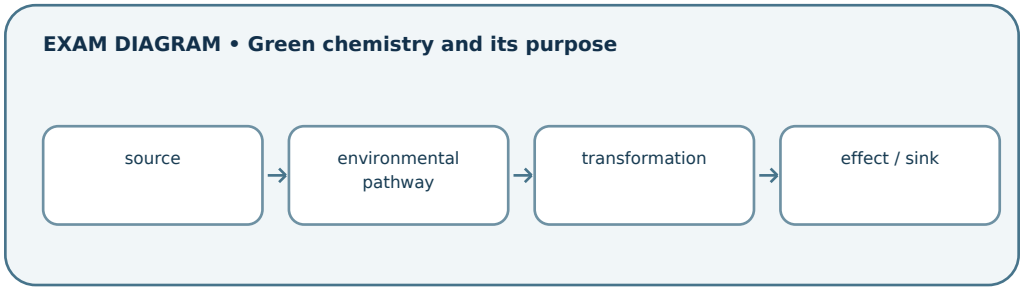
Answer: A. Green chemistry prevents or minimizes hazards at the design stage rather than relying only on later treatment.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.19 The 12 principles of green chemistry

Core concept

The textbook presents twelve principles, including waste prevention, atom economy, less hazardous synthesis, designing safer chemicals, safer solvents, energy efficiency, renewable feedstocks, reducing derivatives, catalysis, design for degradation, real-time analysis for pollution prevention and inherently safer chemistry. Learn the principle name and a one-sentence explanation or example for each.

Remember

Prevention • atom economy • safer synthesis • safer products • safer solvents • energy efficiency • renewable feedstocks • fewer derivatives • catalysis • degradation • real-time analysis • accident prevention

The 12 principles of green chemi	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain any four principles of green chemistry with examples.

B. Which principle focuses on designing products to break down into harmless substances after use?

5.19 • Exam diagram & MCQ practice

Key exam idea: The twelve principles include waste prevention, atom economy, safer synthesis, energy efficiency and design for degradation.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which principle focuses on designing products that break down into harmless substances after use?

A. design for degradation
B. atom economy
C. catalysis
D. real-time analysis

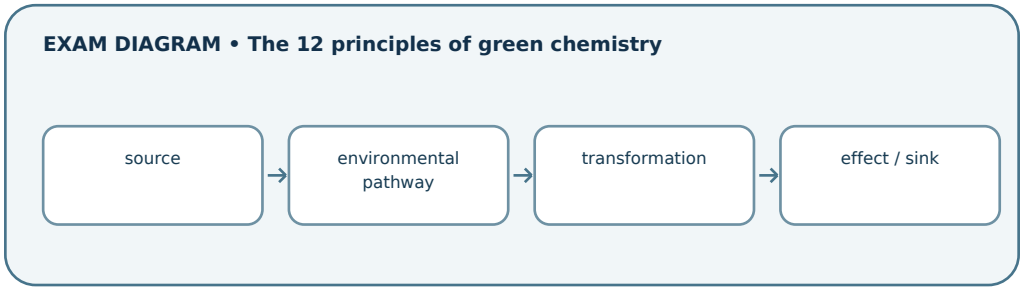
Answer: A. Design for degradation aims for chemical products to break down into harmless substances after their intended use.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



5.20 Cleaner production

Core concept

Cleaner production seeks to improve resource efficiency and reduce waste, energy use, toxicity and unwanted by-products during production. The textbook emphasizes renewable raw materials, less toxic feedstocks and products that degrade more readily. An ideal process balances safety, simplicity, selectivity, high yield, energy efficiency and reduced waste.

Remember

Efficient process + safer inputs + less waste

Cleaner production	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. State the goals of cleaner production in chemistry.

B. Suggest two changes that could make an industrial chemical process cleaner.

5.20 • Exam diagram & MCQ practice

Key exam idea: Cleaner production continuously improves processes to use resources efficiently and reduce waste, emissions and risks.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which action is an example of cleaner production?

A. recovering and reusing process water
B. releasing untreated effluent
C. increasing raw-material losses
D. using more energy for the same output

Answer: A. Water recovery reduces resource consumption and wastewater generation at the source.

Question 2. In the diagram, which labelled step comes immediately after “source”?

A. environmental pathway
B. transformation
C. effect / sink
D. source

Answer: A. The schematic is arranged in sequence: source → environmental pathway → transformation → effect / sink.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.

